

Darwin Smith

San Diego, CA | darwinlls321@gmail.com | [linkedin.com/in/darwin-smith](https://www.linkedin.com/in/darwin-smith) | darwinls.github.io

EDUCATION

University of California, San Diego

B.S. in Mathematics and Computer Science

San Diego, CA

June 2026

EXPERIENCE

Software Engineer & Product Designer

Nov 2025 – Present

Veraflux (veraflux.org), self-founded, pre-launch health-research product

San Diego, CA

- Built Veraflux solo: it writes cited supplement reports from PubMed studies for a given age, sex, and goal, and every change ships through 25 CI check scripts (React/TypeScript, Node/Express, FastAPI, PostgreSQL, Stripe).
- **LLM pipeline (RAG):** 13 stages per uncached report on Gemini and Claude. Parallel efficacy and safety searches hand selectors a median 152 and 134 candidates; one report cited 52 of 229 scanned, none from outside.
- **Label scanning:** a single Gemini vision call (\$0.0017) reads a supplement label, including proprietary-blend amounts. Each ingredient gets a cited brief costing \$0.018, about 1/24th of a full report, cached for every user.
- **Cost tracking:** logged the cost of every model call, found a fully uncached report costs \$0.43, and matched a day's Anthropic invoice exactly. Limited free-tier reports to 165 cache keys no matter how many people sign up.
- **Payments:** quota claims are one atomic SQL statement with refunds, Stripe's 9 webhook types idempotent, and 25 CI check scripts, one keeping all 92 async routes wrapped after 46 hung instead of erroring.
- **Analytics:** built a self-hosted BI suite inside the product's own Postgres: 42 endpoints behind 18 admin pages, a 37-event schema with per-event PII allowlists, and hand-written funnel and retention SQL.
- **Safe deploys:** a version check stops a server deploy from regenerating every cached report before the pipeline can build the new version. The API also sheds load before it charges quota or money.
- **LLM deadlines:** a request's time budget starts when it arrives. Each Claude call and retry gets only the time left, and work queued past its budget is refused before it costs anything.
- **Security:** closed a blind-SSRF path in PubMed XML parsing with a host allowlist, reachable only via NCBI. Its first version missed an NIH subdomain, breaking retrieval; failures now raise their own error.
- **Reproducible schema:** production could not be rebuilt from its migrations. Wrote baseline migrations, replayed the whole history on fresh staging, and confirmed columns and indexes byte-identical to production.
- Pre-rendered 45 public routes with headless Chromium and generated a 237-URL sitemap for search.
- **Frontend:** report scrolling re-rendered the entire report every frame, about 25,500 regex runs per render; fixed it. Also built the component that turns statistics and terms in AI text into accessible popovers.

Test Engineer

Nov 2023 – Apr 2026

SiteHive

San Diego, CA

- Ran US QA and final validation for IoT deployments as technical contact for major enterprise clients.
- Diagnosed firmware and network faults on Hexanode devices with TeraTerm and STM32CubeProgrammer, and tightened validation around recurring hardware defects.

PROJECTS

Club Triton (Collection Strategy Game) | JavaScript, IndexedDB, GitHub Actions

Apr 2025 – June 2025

- Built the game loop and deck-building system (IndexedDB) and merged team work through code reviews.
- Set up CI/CD in GitHub Actions that ran builds, ESLint, and Playwright regression tests on every push.

Global Conflict & Defense Stock Analysis | Python, Pandas, Seaborn

Sep 2025 – Dec 2025

- Merged 20+ years of stock and conflict data; found $r = 0.67$ between conflict deaths and defense valuations.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java, C/C++, C#, Bash

AI & Data: LLM APIs (Gemini, Claude), retrieval-augmented generation (RAG), multimodal vision, structured output, prompt engineering, Batch API, Pandas, NumPy, Scikit-Learn

Frontend: React, Vite, HTML/CSS, Framer Motion

Backend & Databases: Node/Express, FastAPI, REST APIs, PostgreSQL (Prisma), Zod, Stripe, Supabase

Cloud, DevOps & Tooling: Docker, Railway, GitHub Actions, AWS, Azure, Git, Playwright, Jest, JUnit